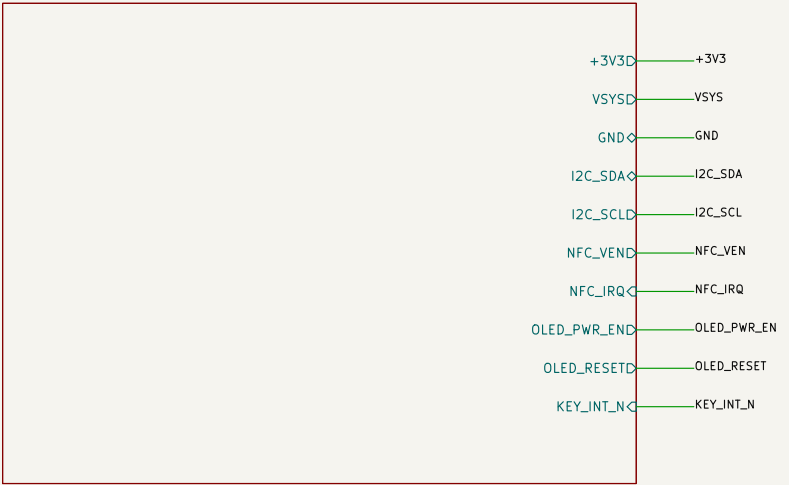
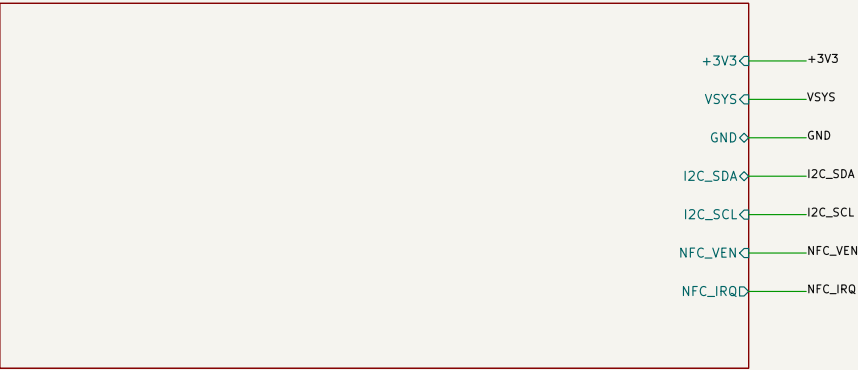


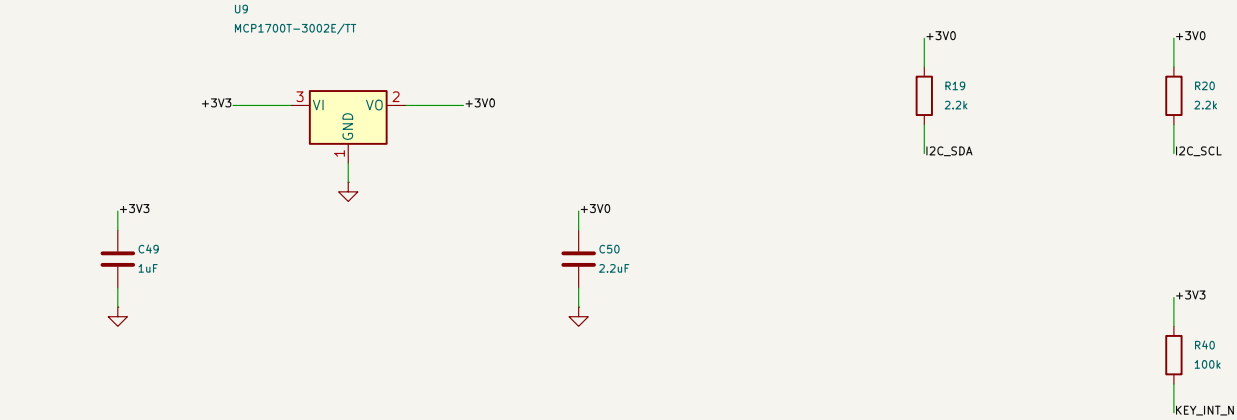
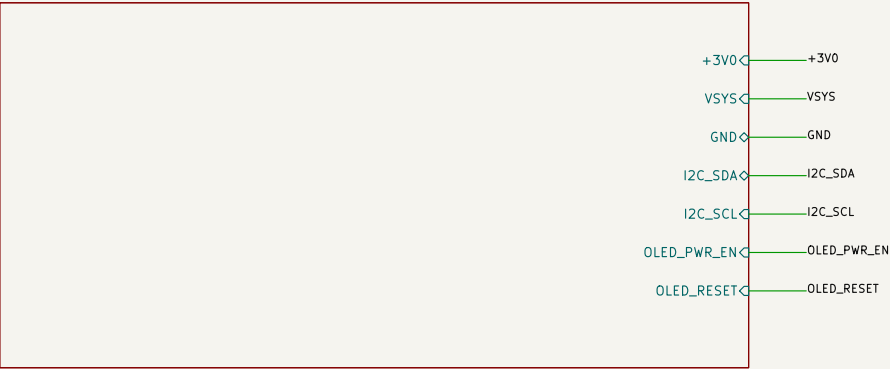
Power + ESP32-C3



NFC / PN7160 / 0x28

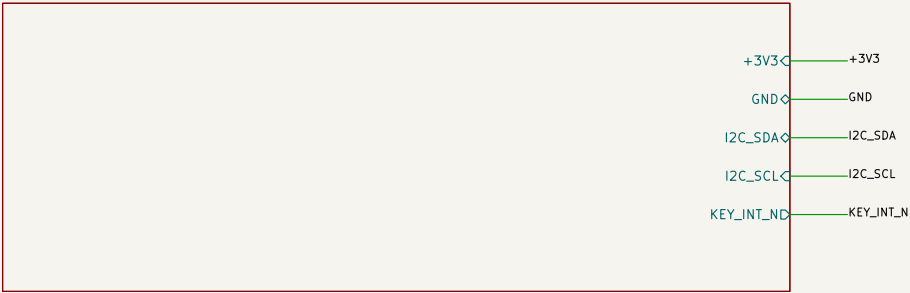


OLED / SSD1309 / 0x3C



3.0 V display logic and common I2C pull-ups.
SSD1309 logic max = 3.3 V; existing buck worst case ~3.39 V.
3.0 V HIGH levels also meet the 3.3 V devices' thresholds.
One populated pair of I2C pull-ups. Bus speed: 100 kHz.
No I2C multiplexer is needed; addresses are distinct.

Keyboard / breakaway / 0x20



J4
MAIN LINK / OPTIONAL

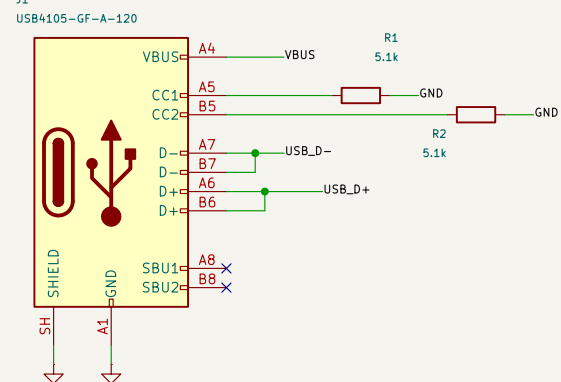


Break line carries five nets only.
Reconnect J4 <-> J5 after separation.
J3 9-pin header; DNP; pins 1 / 9 NC, center 7 active.
Mechanical break line is deferred to PCB layout.

NFC / OLED power comes from VSYS; digital logic from regulated rails.
USB prototype: user-confirmed 5 V / >=1.5 A supply; ~1.10 A estimate.
Battery performance / low-battery improvements deferred this iteration.
OLED: supplied 24-contact CON24 electrical pinout; verify actual flex fit.
RF matching and crystal load selected; final tuning needs measurement.

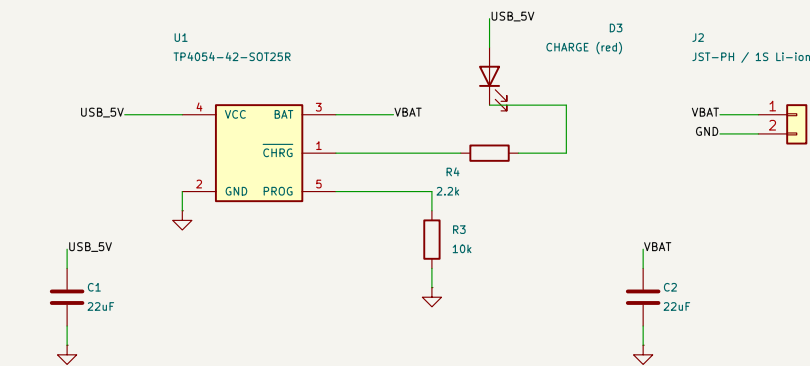
NUCULA v2 / USB-C, Li-ion power and ESP32-C3

01 USB 2.0 device / 5 V sink



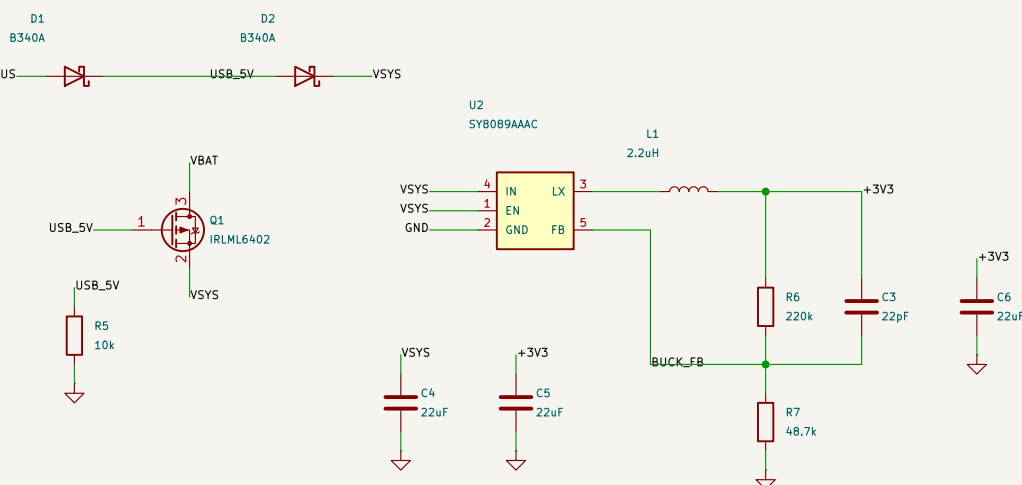
CC1 / CC2: independent 5.1k Rd. No USB-PD.
Route D+/D- as a 90 ohm differential pair.
Place U4 at connector; R12/R13 at module.
Prototype source: 5 V USB-C, at least 1.5 A.
Approx. 1.10 A planning budget; test startup/inrush.

02 4.2 V / 100 mA charger



OPTIONAL battery test; USB works with J2 empty.
J2: JST-PH 2.0mm; pin 1 = BAT+; pin 2 = GND.
100mA charge; protected 1S / 4.2V pack required.
Battery performance deferred; check polarity before use.

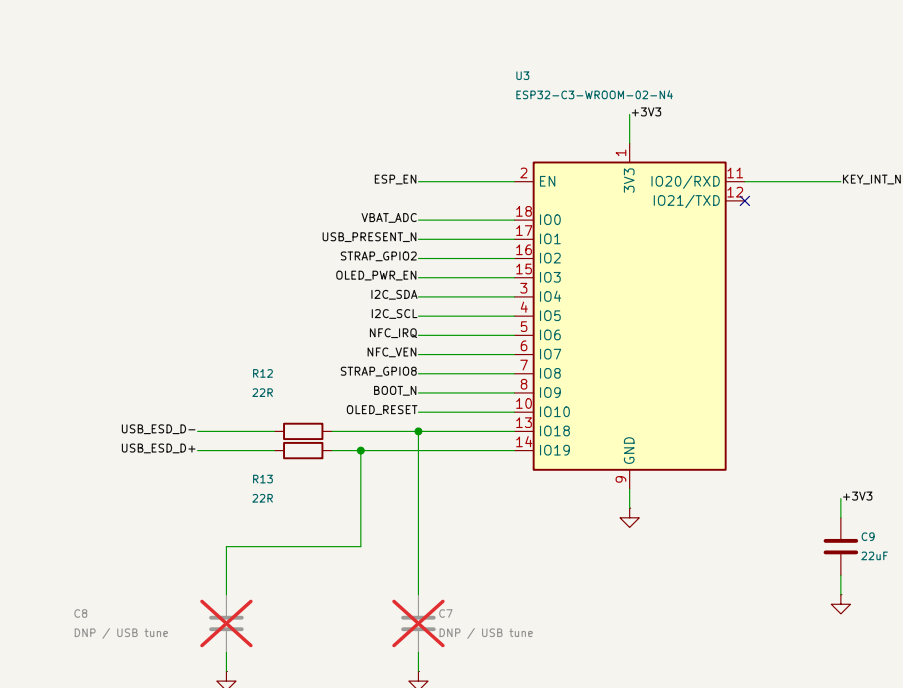
03 USB-priority supply / 3.31 V buck



Q1 drain = battery, source = VSYS.
Body diode points VBAT $\rightarrow$ VSYS.
USB load bypasses charge-current sensing.

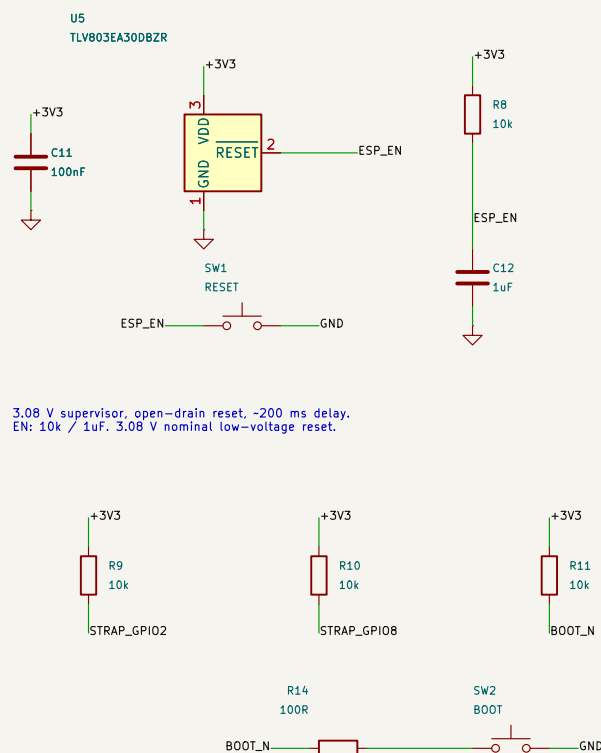
VOUT = 3.31 V; R6 / R7: 0.1% precision
500 mA rail design target; validate transients.
Buck only: low battery causes dropout / reset.
C5/C6: 22uF each, 1206 / 25V; bias curve checked.

04 ESP32-C3-WROOM-02-N4 / native USB



USB D- = GPIO18; D+ = GPIO19. Native USB.
I2C: GPIO4 SDA / GPIO5 SCL; use 100 kHz.
GPIO6 NFC IRQ; GPIO7 VEN; GPIO3 OLED power.
GPIO10 OLED reset; GPIO20 keyboard interrupt.
GPIO21 is reserved; straps GPIO2/8/9 unchanged.

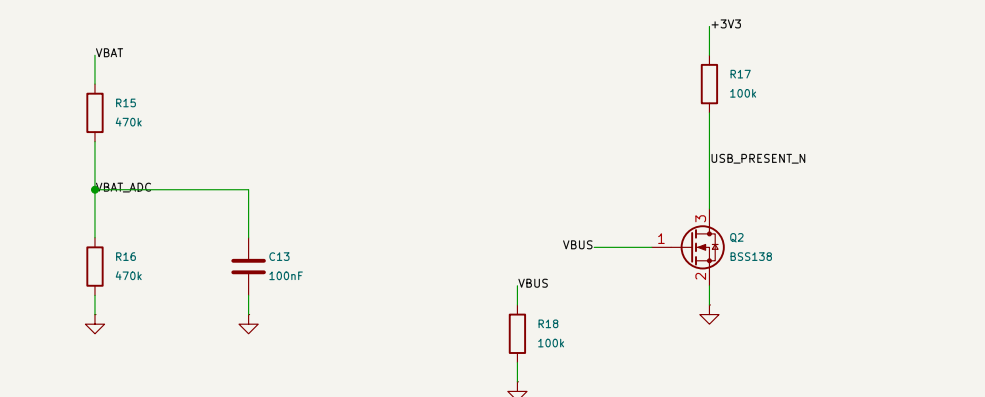
05 Reset, undervoltage and boot straps



3.08 V supervisor, open-drain reset, ~200 ms delay.
EN: 10k / 1uF. 3.08 V nominal low-voltage reset.

Hold BOOT; tap RESET to enter ROM download mode.
GPIO8 must stay high while GPIO9 is low.
No large capacitor on GPIO9.

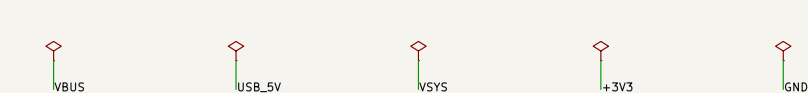
06 Battery and USB presence sensing



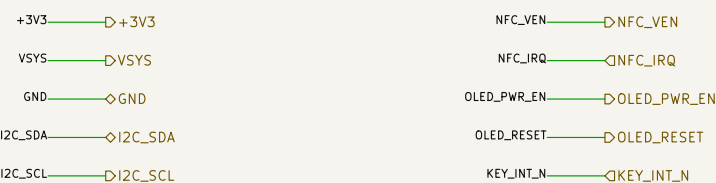
VBAT_ADC = VBAT / 2, max -2.15 V incl. tolerances.
ADC1 GPIO0: use calibrated 11/12 dB attenuation.
Divider load -4.5 uA; allow >=120 ms to settle.

USB_PRESENT_N: GPIO1 LOW when VBUS is present.
Firmware must disable USB attach when VBUS is absent.
Q2 isolates 5 V VBUS from the 3.3 V GPIO supply.

USB PROTOTYPE: dedicated 5 V / ≥ 1.5 A source.
Battery runtime and low-battery changes deferred.
Test inrush, load steps and diode temperature.
C1/C2/C4: 22uF / 25V / 1210; bias curve checked.
Reference: Olimex ESP32-C3-DevKit-Lipo Rev C.
See docs/component-refinements.md for calculations.

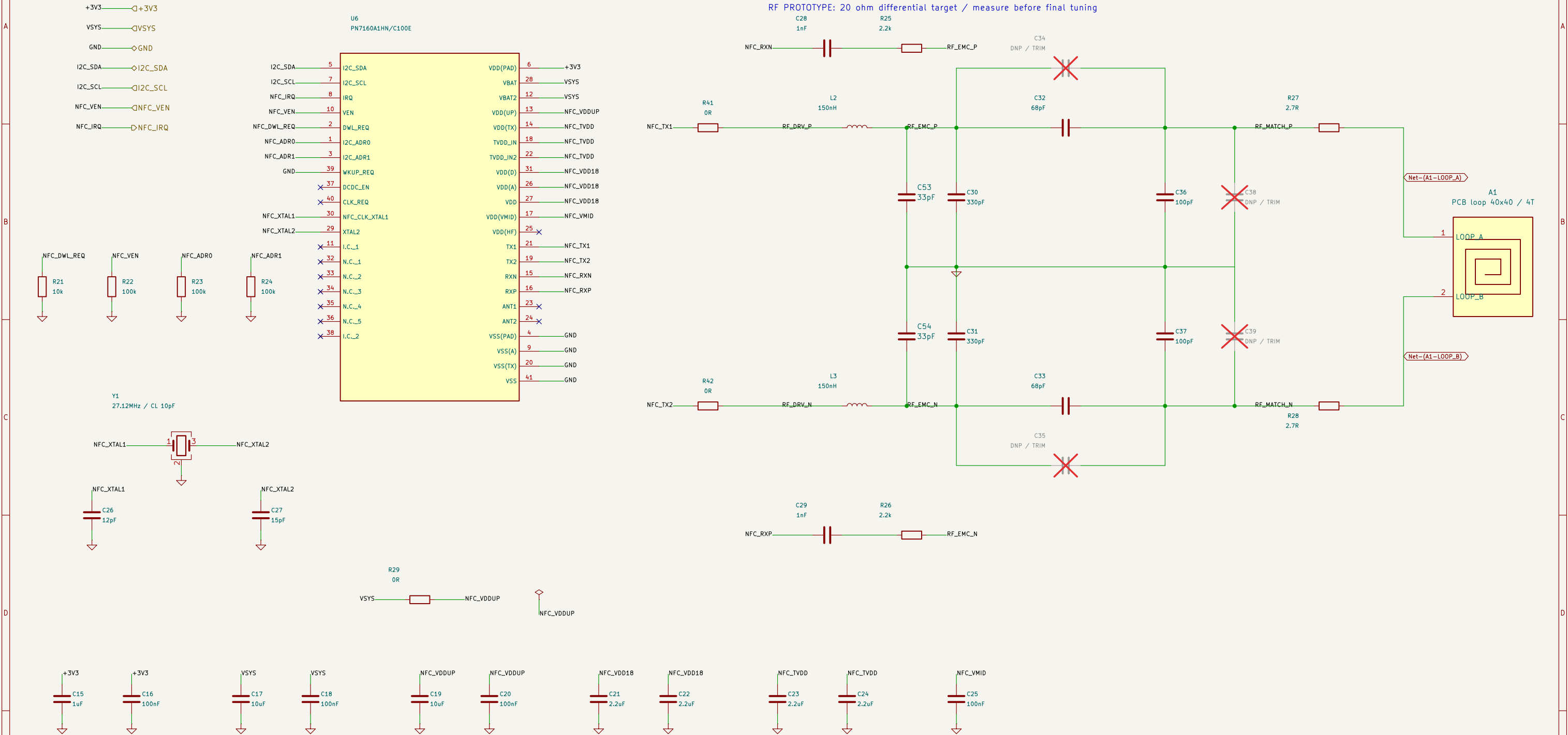


ERC source declarations



Sheet: /Power + ESP32-C3/ File: power-mcu.kicad_sch		
Title: Nucula v2 - USB-C, battery power and ESP32-C3		
Size: A3	Date: 2026-09-20	Rev: B-draft
KiCad E.D.A. 10.0.6		Id: 2/5

PN7160 / I2C host + balanced NFC antenna matching

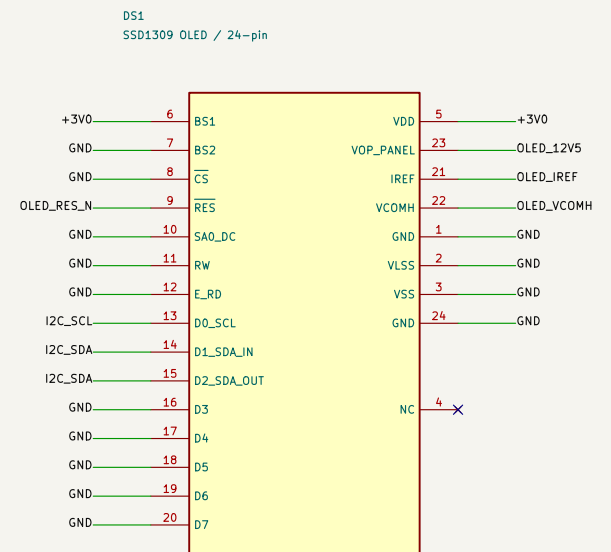
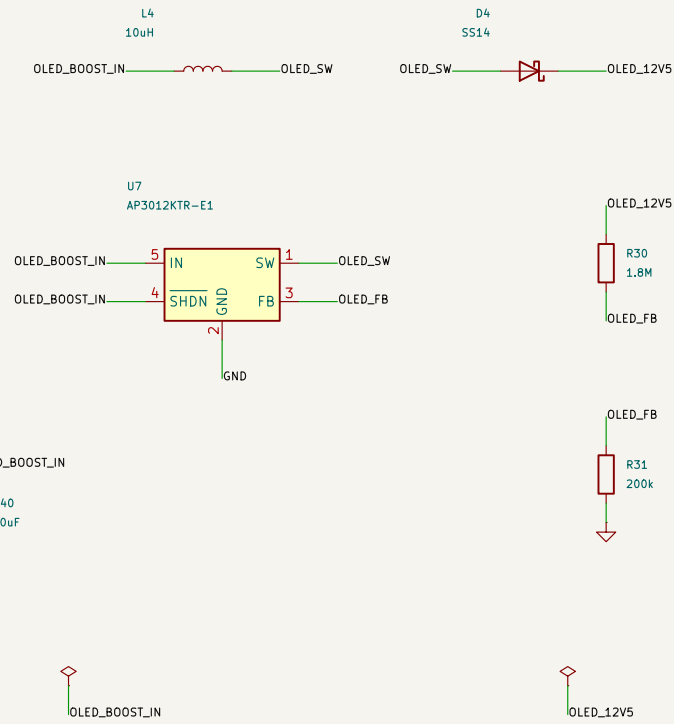
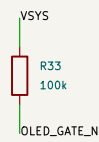
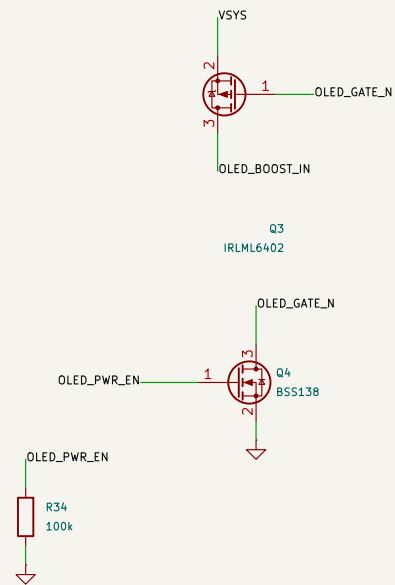
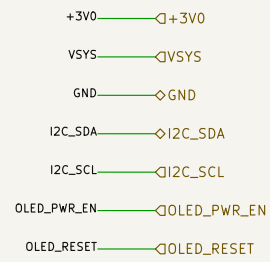


Pin 14 -> 18 + 22: internal TX-LDO output, not an external supply.
Pin 31 -> 26 + 27: internal 1.8 V rail; decouple only.
VSYS powers VBAT/VBAT2 and VDD_UP; VDD_PAD uses +3V3.
ADR1:0 = 00 => 0x28. WKUP_REQ low: wake over I2C.
VEN defaults low. DWL_REQ test access via its labelled net.
ANT1/ANT2/VDHF unused: configure normal RF detection.
Y1: NDK CS06346 27.12 MHz, CL 10pF; C26=12pF / C27=15pF.
C21 at pin 26, C22 at 27; C23 at 18, C24 at 22: 2.2uF each.

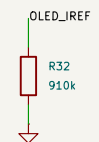
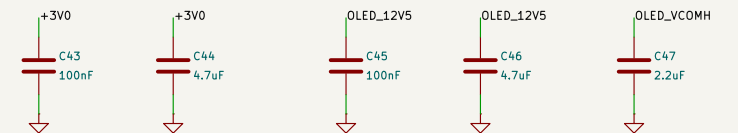
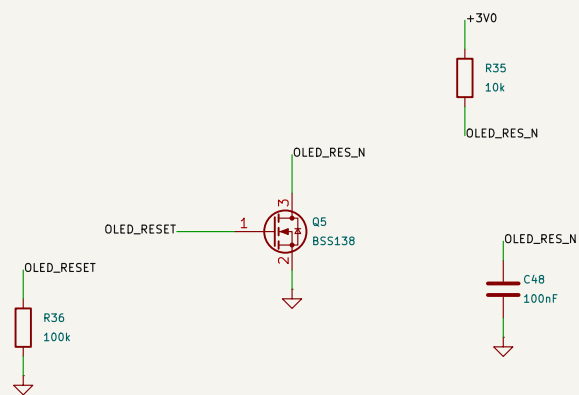
A1: 40x40 mm, 4T, 0.50/0.30 mm; estimated 1.56 uH. No GND tap.
C34/C35, C38/C39: paired 0805 COG tuning pads, DNP.
C30/C31 + C53/C54: 330 + 33 pF = 363 pF EMC shunt per leg.
Lift R41/R42 and R25/R26 to isolate RF tree for passive VNA tests.
Lift R27/R28 for bare-coil measurement. NEVER power a connected VNA.
Start TVDD <= 3.3 V: 20 ohm target trades field strength for current margin.
See docs/nfc-antenna.md for calculations, layout rules and tuning sequence.

OLED / verified CON24 breakout pinout

Switched OLED supply / default OFF



Reset isolation to 3.0 V logic

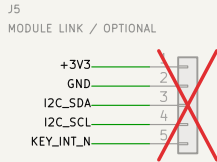
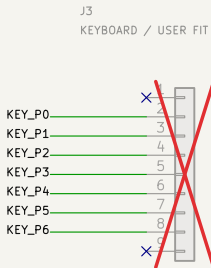
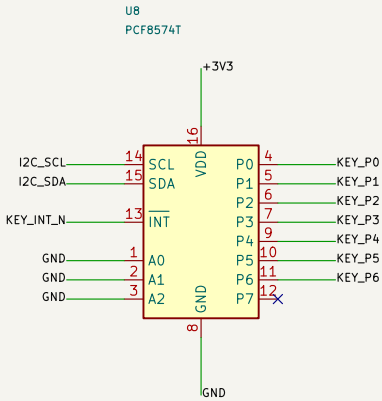
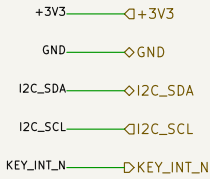


Sequence: logic stable -> reset pulse >=3us -> enable boost -> AFh.
Shutdown: AEh -> disable boost; keep logic powered.
Q3 removes the boost diode feed-through path when OFF.
Do not actively short OLED VCC/VDD to ground (SSD1309 guidance).
12.5 V nominal; retained 1.8M/200k and 910k IREF starting values.
Final glass current, contrast and boost load require validation.

24 contacts: supplied CON24 breakout schematic.
13=SCL; 14+15=SDA; 9=RES#; 23=VOP.
1/2/3/24=GND; 4=NC; 5/6=3.0V.
DS1: FH12A-24S top contact socket only.
NFP1309-02Y: 0.50mm pitch, 0.30mm flex.
Mounted tip contacts face away from PCB.
SSD1309 I2C 0x3C, shared pull-ups 3.0V.

Sheet: /OLED / SSD1309 / 0x3C/ File: oled.kicad_sch	
Title: SSD1309 OLED glass interface and 12.5 V boost	
Size: A3	Rev: B-draft
Date: 2026-09-20	Id: 4/5
KiCad E.D.A. 10.0.6	

BREAKAWAY KEYBOARD / PCF8574T / 0x20



Place every component on this sheet on the detachable PCB section.
Only +3V3, GND, SDA, SCL and INT_N may cross the break line.
Keep connector pads/components clear of tabs, score and fracture area.
J4 (main board) <-> J5 (module): straight-through five-wire reconnect.
J3 is the requested 9-pin keyboard header and is DNP / user fitted.

100 kHz maximum. A2:A0 = 000, address 0x20 (PCF8574T).
J3: pins 1 / 9 NC; pins 2..8 = P0..P6, U8 P7 unused.
Center seven preserve old board J2.1..7 -> P0..P6 order.
Common 9-position keypad: rows P1/P6/P5/P3; columns P2/P0/P4.
Write 1 to release inputs; scan one column low, debounce in firmware.
R38/R39 stay DNP on shared bus; see docs/keyboard-interface.md.

Sheet: /Keyboard / breakaway / 0x20/
File: keyboard.kicad_sch

Title: Detachable PCF8574T keyboard interface

Size: A3
Date: 2026-09-20

Rev: B-draft
Id: 5/5